



ENDOCRINOLOGY

Pituitary Gland - 1

OBJECTIVES

- Understand the classification of diseases of pituitary gland & hypothalamus.
- Know the classical symptoms of hypopituitarism .
- Know how to treat hypopituitarism.

The AIM of this lecture is:

1. Understand the classification of diseases of pituitary gland & hypothalamus.
2. Know the classical symptoms of hypopituitarism .
3. Know how to treat hypopituitarism.

By the end of this lecture the student will be able to:

1. Know the classification of pituitary diseases.
2. Be able to diagnose & treat hypopituitarism.

The pituitary gland is enclosed in the sella turcica and bridged over by a fold of dura mater called the diaphragma sellae, with the sphenoidal air sinuses below and the optic chiasm above. The cavernous sinuses are lateral to the pituitary fossa and contain the 3rd, 4th and 6th cranial nerves and the internal carotid arteries.

The gland is composed of two lobes, anterior and posterior, and is connected to the hypothalamus by the infundibular stalk, which has portal vessels carrying blood from the median eminence of the hypothalamus to the anterior lobe and nerve fibres to the posterior lobe. By far **the most common disorder is an adenoma of the anterior pituitary gland**.

20.50 Classification of diseases of the pituitary and hypothalamus		
	Primary	Secondary
Non-functioning tumours	Pituitary adenoma Craniopharyngioma Metastatic tumours	
Hormone excess		
Anterior pituitary Hypothalamus and posterior pituitary	Prolactinoma Acromegaly Cushing's disease Rare TSH-, LH- and FSH-secreting adenomas Syndrome of inappropriate antidiuretic hormone (SIADH, p. 624)	Disconnection hyperprolactinaemia
Hormone deficiency		
Anterior pituitary Hypothalamus and posterior pituitary	Hypopituitarism Cranial diabetes insipidus	GnRH deficiency (Kallmann syndrome)
Hormone resistance	Growth hormone resistance (Laron dwarfism) Nephrogenic diabetes insipidus	

(FSH = follicle-stimulating hormone; GnRH = gonadotrophin-releasing hormone; LH = luteinising hormone; TSH = thyroid-stimul)

- Tests for Hormones .
- Imaging .
- Biopsy !.



20.51 How to investigate patients with suspected pituitary hypothalamic disease

Identify pituitary hormone deficiency

ACTH deficiency

- Short ACTH stimulation test (see [Box 20.42](#))
- Insulin tolerance test (see [Box 20.55](#)): only if there is uncertainty in interpretation of short ACTH stimulation test (e.g. acute presentation)

LH/FSH deficiency

- In the male, measure random serum testosterone, LH and FSH
- In the pre-menopausal female, ask if the menses are regular
- In the post-menopausal female, measure random serum LH and FSH (FSH normally >30 IU/L and LH >20 IU/L)

TSH deficiency

- Measure random serum T_4
- Note that TSH is often detectable in secondary hypothyroidism

Growth hormone deficiency

Only investigate if growth hormone replacement therapy is being contemplated

- Measure immediately after exercise
- Consider other stimulatory tests (see [Box 20.54](#))

Cranial diabetes insipidus

Only investigate if patient complains of polyuria/polydipsia, which may be masked by ACTH or TSH deficiency

- Exclude other causes of polyuria with blood glucose, potassium and calcium measurements
- Water deprivation test (see [Box 20.60](#)) or 5% saline infusion test

Identify hormone excess

- Measure random serum prolactin
- Investigate for acromegaly (glucose tolerance test) or Cushing's syndrome if there are clinical features

Establish the anatomy and diagnosis

- Consider visual field testing
- Image the pituitary and hypothalamus by MRI or CT

(ACTH = adrenocorticotrophic hormone; CT = computed tomography; FSH = follicle stimulating hormone; LH = luteinising hormone; MRI = magnetic resonance imaging; TSH = thyroid-stimulating hormone)

pathway.

- The resulting visual field defect can be documented using a Goldmann perimetry chart or with an automated perimetry method .
- MRI reveals ‘abnormalities’ of the pituitary gland in as many as 10% of ‘healthy’ middle-aged people.
- It should therefore be performed only if there is a clear biochemical abnormality or if a patient presents with clinical features of pituitary tumour
- A pituitary tumour may be classified as either a **macroadenoma** (>10 mm diameter) or a **Microadenoma** (<10 mm diameter).
- Surgical biopsy is usually only performed as part of a therapeutic operation.
- Conventional histology identifies tumours as chromophobe (usually non-functioning), acidophil (typically prolactin- or growth hormone- secreting) or basophil (typically ACTH-secreting); immunohistochemistry may confirm their secretory capacity but is poorly predictive of growth potential of the tumour

Presenting problems in hypothalamic and pituitary disease

- Younger women with pituitary disease most commonly present with secondary amenorrhoea or galactorrhoea (in hyperprolactinaemia).
- Post-menopausal women and men of any age are less likely to report symptoms of hypogonadism and so are more likely to present late with larger tumours causing visual field defects.
- The presentation and management of pituitary and hypothalamic disease may be affected by old age .
- Nowadays, many patients present with the incidental finding of a pituitary tumour on a CT or MRI scan.

ستاجير

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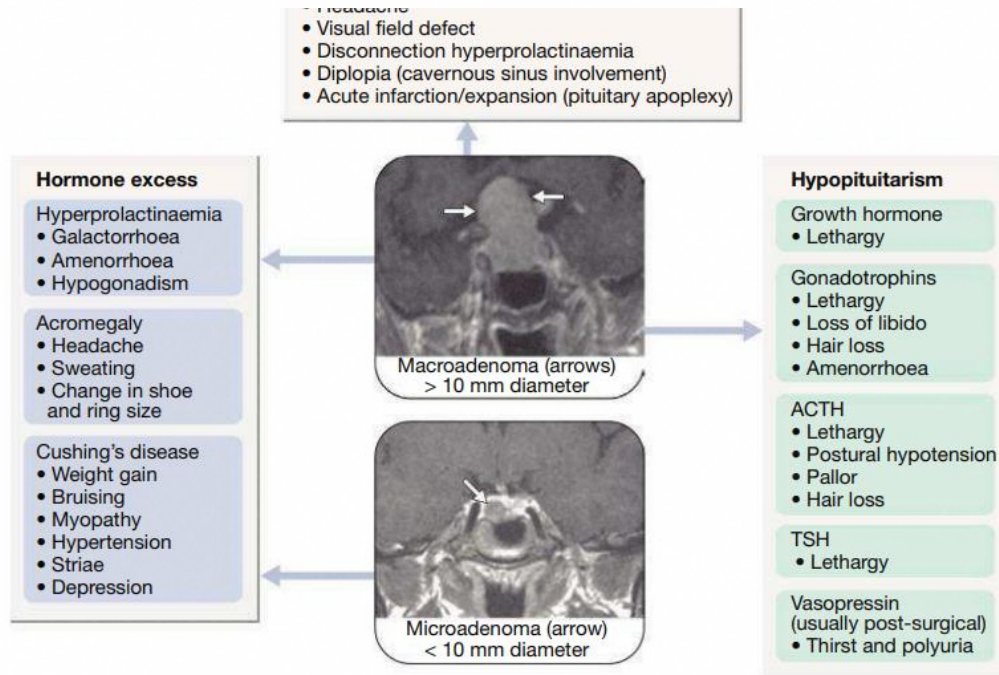


Fig. 20.29 Common symptoms and signs to consider in a patient with suspected pituitary disease. (ACTH = adrenocorticotrophic hormone; TSH = thyroid-stimulating hormone)

Hypopituitarism

- Hypopituitarism describes combined deficiency of any of the anteriorpituitary hormones.
- The clinical presentation is variable and depends on the underlying lesion and the pattern of resulting hormone deficiency.
- The most common cause is a pituitary macroadenoma but there are other causes!.

20.53 Causes of anterior pituitary hormone deficiency	
Structural	
• Primary pituitary tumour • Adenoma* • Carcinoma (exceptionally rare) • Craniopharyngioma* • Meningioma* • Secondary tumour (including leukaemia and lymphoma)	• Chordoma • Germinoma (pinealoma) • Arachnoid cyst • Rathke's cleft cyst • Haemorrhage (apoplexy)
Inflammatory/infiltrative	
• Sarcoidosis • Infections, e.g. pituitary abscess, tuberculosis, syphilis, encephalitis	• Lymphocytic hypophysitis • Haemochromatosis • Langerhans cell histiocytosis
Congenital deficiencies	
• GnRH (Kallmann syndrome)* • GHRH*	• TRH • CRH
Functional*	
• Chronic systemic illness • Anorexia nervosa	• Excessive exercise
Other	
• Head injury* • (Para)sellar surgery* • (Para)sellar radiotherapy*	• Post-partum necrosis (Sheehan syndrome) • Opiate analgesia
*The most common causes of pituitary hormone deficiency. (CRH = corticotrophin-releasing hormone; GHRH = growth hormone-releasing hormone; GnRH = gonadotrophin-releasing hormone; TRH = thyrotrophin-releasing hormone)	

- The presentation is highly variable. For example, following radiotherapy to the pituitary region, there is a characteristic [sequence of loss of pituitary hormone secretion](#).
- [Growth hormone secretion is often the earliest to be lost](#).
- In adults, this produces lethargy, muscle weakness and increased fat mass but these features are not obvious in isolation.
- Next, gonadotrophin (LH and FSH) secretion becomes impaired with loss of libido in the male and oligomenorrhoea or amenorrhoea in the female.
- Later, in the male there may be gynaecomastia and decreased frequency of shaving.
- In both sexes, axillary and pubic hair eventually become sparse or even absent and the skin becomes characteristically finer and wrinkled.
- Chronic anaemia may also occur.
- The next hormone to be lost is usually ACTH, resulting in symptoms of [cortisol insufficiency](#) (Including postural hypotension and a dilutional hyponatraemia).
- [In contrast to primary adrenal insufficiency, angiotensin II-dependent zona glomerulosa function is not lost and hence aldosterone secretion maintains normal plasma potassium](#).
- In contrast to the pigmentation of Addison's disease due to high levels of circulating ACTH acting on the skin melanocytes, a striking degree of pallor is usually present
- [Finally, TSH secretion is lost with consequent secondary hypothyroidism](#).
- This contributes further to apathy and cold intolerance.
- In contrast to primary hypothyroidism, [frank myxedema is rare, presumably because the thyroid retains some autonomous function](#).
- The onset of all of the above symptoms is notoriously insidious.
- However, patients sometimes [present acutely](#) unwell with glucocorticoid deficiency. This may be precipitated by a mild infection or injury, or may occur secondary to [pituitary apoplexy](#) !



Management

Cortisol replacement:

Hydrocortisone should be given if there is ACTH deficiency , it is the most important step in the management.

Mineralocorticoid replacement is not required because sodium depletion is less likely to be present as there is still mineralocorticoid activity due to the renin–angiotensin– aldosterone system (which is lost in primary adrenal insufficiency).

Levothyroxine 50–150 µg once daily should be given , Unlike in primary hypothyroidism, measuring TSH is not helpful in adjusting the replacement dose because patients with hypopituitarism often secrete glycoproteins which are measured in the TSH assays but are not bioactive.

The aim is to maintain serum T4 in the upper part of the reference range.

It is dangerous to give thyroid replacement in adrenal insufficiency without first giving glucocorticoid therapy, since this may precipitate adrenal crisis.

Sex hormone replacement

This is indicated if there is gonadotrophin deficiency in women under the age of 50 and in men to restore normal sexual function and to prevent osteoporosis.

Growth hormone replacement

Growth hormone (GH) is administered by daily subcutaneous self-injection to children and adolescents with GH deficiency and, until recently, was discontinued once the epiphyses had fused.

patients feel better, and have objective improvements in their fat:muscle mass ratio and other metabolic parameters, if they are also given GH replacement.

Treatment with GH may also help young adults to achieve a higher peak bone mineral density.

The principal side-effect is sodium retention, manifest as peripheral oedema or carpal tunnel syndrome if given in excess.

For this reason, GH replacement should be started at a low dose, with monitoring of the response by measurement of serum IGF-1.

Pituitary tumours produce a variety of mass effects, depending on their size and location, but also present as incidental findings on CT or MRI, or with hypopituitarism.

Most intrasellar tumours are pituitary macroadenomas (most commonly non-functioning adenomas, whereas suprasellar masses may be craniopharyngiomas). The most common cause of a parasellar mass is a meningioma.

Clinical assessment of pituitary tumours:

- A common but non-specific presentation is with headache, which may be the consequence of stretching of the diaphragma sellae.
- Although the classical abnormalities associated with compression of the optic chiasm are bitemporal hemianopia or upper quadrantanopia, any type of visual field defect can result from suprasellar extension of a tumour because it may compress the optic nerve (unilateral loss of acuity or scotoma) or the optic tract (homonymous hemianopia). Optic atrophy may be apparent on ophthalmoscopy.
- Lateral extension of a sellar mass into the cavernous sinus with subsequent compression of the 3rd, 4th or 6th cranial nerve may cause diplopia and strabismus, but in anterior pituitary tumours this is an unusual presentation.
- Occasionally, pituitary tumours infarct or there is bleeding into cystic lesions.
- This is termed 'pituitary apoplexy' and may result in sudden expansion with local compression symptoms and acute-onset hypopituitarism.
- Nonhaemorrhagic infarction can also occur in a normal pituitary gland; predisposing factors include catastrophic obstetric haemorrhage (Sheehan syndrome), diabetes mellitus and raised intracranial pressure

A. Patients suspected of having a pituitary tumour should undergo MRI.

If MRI is not available CT may be used but is inferior for assessment of pituitary disease and will not show the majority of small pituitary tumours.

B. Hormonal assessment.

Management

- Urgent treatment is required if there is evidence of pressure on visual pathways.
- The chances of recovery of a visual Field defect are proportional to the duration of symptoms, with full recovery unlikely if the defect has been present for longer than 4 months.
- In the presence of a sellar mass lesion, it is crucial that serum prolactin is measured before emergency surgery is performed.
- If the prolactin is over 5000 mU/L, then the lesion is likely to be a macroprolactinoma and to respond to a dopamine agonist with shrinkage of the lesion, making surgery unnecessary.
- Associated hypopituitarism should be treated.
- Most operations on the pituitary are performed using the trans-sphenoidal approach via the nostrils, while transfrontal surgery via a craniotomy is reserved for suprasellar tumours.
- It is uncommon to be able to resect lateral extensions into the cavernous sinuses.
- All operations on the pituitary carry a risk of damaging normal endocrine function; this risk increases with the size of the primary lesion.
- Pituitary function should be retested 4–6 weeks following surgery, primarily to detect the development of any new hormone deficits.
- Rarely, the surgical treatment of a sellar lesion can result in recovery of hormone secretion that was deficient pre-operatively.

i 20.56 Therapeutic modalities for functioning and non-functioning hypothalamic and pituitary tumours

	Surgery	Radiotherapy	Medical	Comment
Non-functioning pituitary macroadenoma	1st line	2nd line	–	
Prolactinoma	2nd line	2nd line	1st line Dopamine agonists	Dopamine agonists usually cause macroadenomas to shrink
Acromegaly	1st line	2nd line	2nd line Somatostatin analogues Dopamine agonists GH receptor antagonists	Medical therapy does not reliably cause macroadenomas to shrink Radiotherapy and medical therapy are used in combination for inoperable tumours
Cushing's disease	1st line	2nd line	2nd line Steroidogenesis inhibitors Pasireotide	Radiotherapy may take many years to reduce ACTH excess and medical therapies may be used as a bridge. Bilateral adrenalectomy may also be considered if the pituitary tumour is not completely resectable
Craniopharyngioma	1st line	2nd line	–	

(ACTH = adrenocorticotrophic hormone; GH = growth hormone)



" لا تنسوا الدعاء لأهالينا في فلسطين "